

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st March 2017

Subject SA5 – Finance

Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) An organic farmer is interested in growing of “Bharat Herbicia”, a medicinal plant which is used in preparation of many Ayurveda medicines. The roots of the plant is used to extract the juice called "Actmed phosphorous" and the amount of Actmed contained in the plant is what determines the grade of the plant. The plant is graded in four categories:

- I.** Grade A – highest Actmed content and used in larger number of medicines including medicines for treating diabetes
- II.** Grade B – medium Actmed content and used in medium number of medicines
- III.** Grade C – low Actmed content and used in few number of medicines
- IV.** Grade D – plants that have either died or likely to die

At any point of time any given plant can move from one category to another other than Category D which is the absorbing state. It takes 3 months for the plant to grow and be ready for cultivation. The probability of changing state is observed at 20% month on month basis.

The farmer has 10,000 acres of land in Kerala and is considered useful for a wide variety of crops all through the year including "Bharat herbicia".

You are hired as a consultant to advice on helping the farmer to make the right decision and helping him to build a sustainable business model.

- i)** List the factors that you will consider in evaluating the proposal of growing "Bharat herbicia" (4)
- ii)** How will you apply RAMP methodology to this business proposal; and describe what all the factors you will consider in each phase of RAMP process? (9)
- iii)** Describe five key risks that you think need to be managed for this business to be successful in the long term? (6)
- iv)** Explain how the matrix capturing the grade of each plant at the end of each month can be useful for project appraisal? (5)
- v)** Explain the approach you will use for appraising the financial feasibility of inventions and innovations; and discuss whether the approach can be applied for appraising this project. (7)
- vi)** Define Liquidity risk? (1)
- vii)** Discuss the sources of liquidity risk for this business; and the ways in which the farmer can manage the liquidity risk. (5)
- viii)** A "venture capitalist" is interested in this business proposal and is willing to invest up to Rs.100 crores over three years in exchange of 40% share in profit and complete management control. Briefly discuss how this business can be evaluated from the standpoint of the venture capitalist; and the exit options that may be available to the venture capitalist assuming that the venture is successful (12)
- ix)** The farmer wants to approach the Ministry of Science and Technology for granting an investment subsidy for this project. Discuss the factors that will be considered by the Ministry before providing an investment subsidy for this project (6)

[55]

Q. 2) Based on the recommendations provided by you, the farmer growing “Bharat Herbicia” [referred to in Question I] has incorporated a limited liability company - Bharat Herbals Ltd (BHL)- in which the venture capitalist has a 40% equity stake. Over a period of time, the company has prospered and has a net worth of over Rs. 500 Crores as on date. BHL has also been actively engaged in encouraging the farmers in the nearby villages in organic farming to grow various crops besides medicinal plants by giving them technical advice and sometime helping them with financial assistance as its social responsibility. Over a period of time it has gained considerable credibility in the local communities.

At this point in time, the country's central bank in order to encourage micro credit, has announced that it would issue licenses for micro-credit focused banks. BHL, given its track record in farming and experience in rural areas, is contemplating to apply for one of the micro finance bank licenses.

You are hired as a consultant to help the company for setting up the bank.

- i)** Identify the factors that the central bank would consider in assessing BHL's eligibility for the license. (7)
- ii)** Briefly describe the key factors which a micro-finance institution should consider while evaluating the credit risk given that the focus of this institution is on providing short term and medium term loans to the farming community and to the small business units located in the rural areas. (7)
- iii)** Explain how the credit risk faced by a micro-finance institution can be reduced using securitization (6)

The central bank has granted BHL license and the bank BHL Rural Bank Ltd (BRB) has been incorporated. The bank has been progressing well. In order to strengthen their capital adequacy, the bank is contemplating to issue bonds to raise capital, which may qualify as Tier-2 capital. The Bank has approached a credit rating agency for rating the bonds.

- iv)** What do you understand by the term “Credit Rating”; and discuss the limitations of credit rating from the standpoint of the users of such ratings. (5)
- v)** Describe the factors the credit rating agency would consider before assigning a rating for the bonds proposed to be issued by BRB (8)

The CEO of BRB has recently attended a seminar on “Enterprise Risk Management for Banks” where the speakers placed considerable emphasis on identifying, assessing and managing interest rate risk, credit risk and operational risk. He wants you to prepare a report on the relevance of interest rate risk and operational risk for a micro-finance institution like BRB.

- vi)** Draft a report describing the sources of interest rate risk and operational risks for BRB and suggest ways for managing these risks. (12)
- [45]**
